INCH-POUND

MIL-M-28787/219B(SH)
AMENDMENT 1
27 November 1991

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

AMPLIFIER, POWER, ANALOG, KEY CODE HEE

This amendment forms a part of Military Specification MIL-M-28787/219B(EC), dated 15 November 1984, and is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 4

TABLE I: Delete and substitute new table I as printed on page 3 of this amendment.

PAGE 5

TABLE I: Delete and substitute new table I as printed on page 4 of this amendment.

PAGE 7

6.5, Nonintegrating bandwidth, class I and class II. Delete ">15" (two places) and substitute "15" (two places).

PAGE 17

TABLE VI: Delete and substitute new table VI as printed on page 5 of this amendment.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:45Z
Check the source to verify that this is the current version before use.

MIL-M-28787/219B(SH)
AMENDMENT 1

PAGE 27

TABLE XIII, Test conditions P1 and P2, Max column: Delete "10" (two places) and substitute "25" (two places).

Preparing activity:
Navy - SH

(Project 5963-N801)

Agent NW

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:45Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25 ± 5° 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	II	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	II	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B	Stress	III	Q	10	15	--	--	--	--	--	--	--	--	sec
C1	Output current limit	III	S	2.4	4.8	2.4	4.8	2.4	4.8	2.4	4.8	2.4	4.8	Vpk
C2	Output current limit	III	S	-4.8	-2.4	-4.8	-2.4	-4.8	-2.4	-4.8	-2.4	-4.8	-2.4	Vpk
D1, D2	Quiescent power supply current	IV	X	0.5	35.0	0.5	35.0	0.5	35.0	0.5	35.0	0.5	35.0	mA
E1	Gain (Gain = 5)													
	Initial	V	X	0 165973	0 167362	0.165809	0.167532	0.165700	0.167644	0.165425	0.167926	0.165508	0.167841	Ratio
	End-of-life	V	X	0 165617	0 167728	0.165453	0 167897	0.165234	0 168123	0.165070	0.168293	0 165070	0.168293	Ratio
E2	Gain (Gain = 6.67)													
	Initial	V	X	0.129812	0.130946	0.129678	0.131084	0.129589	0.131176	0.127365	0.131406	0.129432	0.131337	Ratio
	End-of-life	V	X	0 129522	0 131245	0.129387	0.131383	0.129209	0.131568	0.129076	0.131706	0.129076	0.131706	Ratio
E3	Gain (Gain = 10)													
	Initial	V	X	0.090496	0 091323	0.090399	0.091424	0.090334	0.091491	0.090171	0.091659	0.090220	0.091608	Ratio
	End-of-life	V	X	0.090285	0.091541	0.090187	0.091642	0.090057	0.091776	0.089960	0.091877	0.089960	0.091877	Ratio
E4	Gain, integrator													
	Initial	V	X	0 090541	0 091393	0 090496	0 091439	0 090496	0.091439	0.090454	0.091485	0.090481	0.091460	Ratio
	End-of-life	V	X	0.090372	0 091550	0.090291	0.091649	0.090291	0.091649	0.090250	0.091693	0.090250	0.091693	Ratio
F1-F8, F11, F12	Frequency response	VI	X	0.97	1.05	0.97	1.05	0.97	1.05	0.97	1.05	0.97	1.05	Vrms
F9	Frequency response	VI	X	0.64	0.76	0.64	0.76	0.64	0.76	0.64	0.76	0.64	0.76	Vrms
F10	Frequency response	VI	X	0.10	0.30	0.10	0.30	0.10	0.30	0.10	0.30	0.10	0.30	Vrms
F13	Frequency response	VI	X	0.93	1.02	0.93	1.02	0.93	1.02	0.93	1.02	0.93	1.02	Vrms
F14	Frequency response	VI	X	0.88	0 97	0 88	0.97	0 88	0.97	0.88	0.97	0.88	0.97	Vrms

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:45Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25 ± 5° 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
F15	Frequency response	VI	X	0.85	0.94	0.85	0.94	0.85	0.94	0.85	0.94	0.85	0.94	Vrms
F16	Frequency response	VI	X	0.69	0.82	0.69	0.82	0.69	0.82	0.69	0.82	0.69	0.82	Vrms
F17	Frequency response	VI	X	0.67	0.80	0.67	0.80	0.67	0.80	0.67	0.80	0.67	0.80	Vrms
F18	Frequency response	VI	X	0.60	0.74	0.60	0.74	0.60	0.74	0.60	0.74	0.60	0.74	Vrms
F19	Frequency response	VI	X	0.50	0.66	0.50	0.66	0.50	0.66	0.50	0.66	0.50	0.66	Vrms
F20	Frequency response	VI	X	0.02	0.04	0.02	0.04	0.02	0.04	0.02	0.04	0.02	0.04	Vrms
G	Noise	VII	Q	0	100	0	100	0	100	0	100	0	100	mVpk
H1	Phase accuracy	VIII	X	181	186	181	186	181	186	181	186	181	186	Deg
H2	Phase accuracy	VIII	X	178	183	178	183	178	183	178	183	178	183	Deg
J	Output voltage swing	IX	X	5.6	9.7	5 6	9.7	5.6	9.7	5.6	9.7	5.6	9.7	Vrms
K	Oscillation	IX	X	◄---------- There shall be no oscillation present ----------►										
L	Input impedance	X	S	2.48	2 52	2.48	2.52	2.48	2.52	2.48	2.52	2.48	2.52	Vrms
M	Output offset voltage	XI	X	-13	13	-13	13	-13	13	-13	13	-13	13	mV
N1, N2	P S capacitance	XII	X	5.4	8.2	5.4	8.2	5.4	8.2	5.4	8.2	5.4	8.2	µF
P1, P2	P. S ripple rejection	XIII	Q	0	25	0	25	0	25	0	25	0	25	mV
Q1, Q2	Diode forward voltage	XIV	S	0.4	1.0	0.4	1.0	0.4	1.0	0.4	1.0	0.4	1.0	V

NOTES:

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (one percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:45Z
Check the source to verify that this is the current version before use.

TABLE VI. Frequency response test.

Test condition	Parameter	Item G10 frequency Hz ± 1%	Connect A to pin	Initial limits 25 ± 5°C, ambient		Unit	Test figure
				Min	Max		
F1	Frequency response	50	Open	0.97	1.05	Vrms	9
F2	Frequency response	60	Open	0.97	1.05	Vrms	9
F3	Frequency response	100	Open	0.97	1.05	Vrms	9
F4	Frequency response	400	Open	0.97	1.05	Vrms	9
F5	Frequency response	800	Open	0.97	1.05	Vrms	9
F6	Frequency response	1000	Open	0 97	1.05	Vrms	9
F7	Frequency response	1500	Open	0.97	1.05	Vrms	9
F8	Frequency response	2000	Open	0.97	1.05	Vrms	9
F9	Frequency response	15000	Open	0.64	0.76	Vrms	9
F10	Frequency response	50000	Open	0.10	0.30	Vrms	9
F11	Frequency response	50	18	0.97	1.05	Vrms	9
F12	Frequency response	60	18	0.97	1.05	Vrms	9
F13	Frequency response	400	18	0.93	1.02	Vrms	9
F14	Frequency response	800	18	0.88	0.97	Vrms	9
F15	Frequency response	1000	18	0.85	0.94	Vrms	9
F16	Frequency response	1590	18	0.69	0.82	Vrms	9
F17	Frequency response	1700	18	0.67	0.80	Vrms	9
F18	Frequency response	2000	18	0.60	0.74	Vrms	9
F19	Frequency response	2500	18	0.50	0.66	Vrms	9
F20	Frequency response	50000	18	0.02	0.04	Vrms	9

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:45Z
Check the source to verify that this is the current version before use.